Media vs. Public Opinion on GLP-1 Weight-Loss Drugs

A Comparative Text-Analytics Study of Ozempic & Wegovy

Vasilis Christopoulos, Hugo Guideau, Saksi Khosla, Mustafa Yousuf, Othmane Zizi
INSY 669 – Text Analytics | McGill University | Winter 2026

1. Introduction

GLP-1 receptor agonists—Ozempic and Wegovy—have sparked extensive online patient discussion and media coverage. This study compares **public discourse** (3,246 Reddit posts + 102 WebMD reviews) with **media discourse** (634 Google News articles) from January–November 2024, asking: (1) How does sentiment differ? (2) What linguistic and thematic gaps exist? (3) Does either stream lead the other temporally?

2. Data & Methodology

Source	**Type**	n	**Method**
Reddit	Public	3,246	Arctic Shift API
WebMD	Public	102	Web scraping
Google News	Media	634	RSS + URL decode

We used a **hybrid** media-text strategy (full article body when available, snippet fallback otherwise). Preprocessing included tokenisation, stopword removal, and lemmatisation. A **length-normalised track** (first 40 tokens) controls for document-length confounds. All ML pipelines use stratified k-fold CV to prevent leakage.

3. Results

3.1 Sentiment Analysis

VADER scores reveal a significant gap: public mean $= +0.121$, media $= -0.137$ ($t = 9.43$, $p < 0.001$, Cohen's $d = 0.44$). Reddit is moderately positive (+0.135), WebMD strongly negative (−0.308), and news negative (−0.137). All pairwise Mann–Whitney U tests are significant ($p < 0.001$).

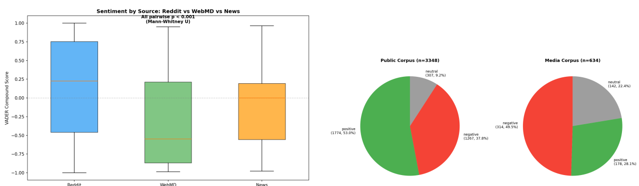

Figure 1: Sentiment by source and label distribution.

3.2 Corpus Comparison & Side Effects

TF-IDF top terms confirm thematic divergence: public emphasises *week, dose, lb, anyone*; media foregrounds *semaglutide, drug, weightloss drug*. Cosine similarity between mean vectors: 0.346 (0.350 normalised).

A targeted side-effect analysis shows public mentions nausea (3.38/1k tokens), diarrhea (1.22), constipation (1.16), and anxiety (0.69) at rates far exceeding media, where most side effects have zero mentions. Gaps persist under normalisation.

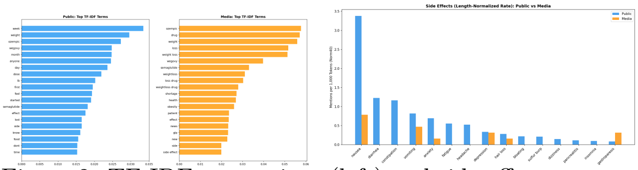

Figure 2: TF-IDF comparison (left) and side-effect rates per 1k tokens (right).

3.3 Classification

Naïve Bayes (α=0.1): **97.6%** CV accuracy. KNN (k=7): **96.7%**. Normalised-track accuracy is nearly identical (97.6% / 96.4%), confirming separability is vocabulary-driven, not length-driven. Top public features: *anyone, day, lb, feeling*; top media: *cnn, healthline, medscape, nbc*.

3.4 Topic Modelling

LDA (5 topics/corpus) reveals public topics centred on dosing, insurance, weight progress, and compounding; media topics on clinical trials, FDA regulation, and Medicare access. K-means (k=2) purity: **84.1%**—strong unsupervised separation.

3.5 Aspect-Based Sentiment

Sentiment was computed across seven aspects. The largest discrepancies:

Aspect	**Pub.**	**Med.**	**Gap**	**Sig.**
Mental health	−0.10	−0.74	0.63	p=.040*
Access	+0.20	−0.35	0.55	p<.001*
Efficacy	+0.14	−0.33	0.46	p<.001*
Cost	+0.16	−0.02	0.18	p=.017*
Side effects	−0.03	−0.20	0.16	p=.666

Media is dramatically more negative on mental health (−0.74 vs. −0.10), likely driven by alarming headlines. A discrepancy logistic-regression classifier achieves 83.7% accuracy.

3.6 Temporal Lead-Lag

Granger causality tests (max lag 4 weeks, 45 observations) find no significant relationship in either direction (Media→Public p=0.37; Public→Media p=0.30). Cross-correlation peaks at lag −7 weeks (r=0.31). The two streams appear to respond to shared events rather than driving each other.

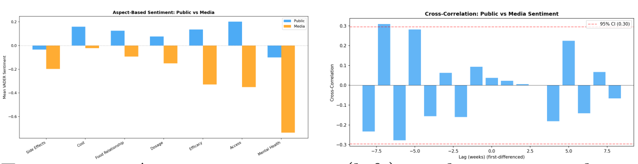

Figure 3: Aspect sentiment (left) and cross-correlation (right).

4. Discussion

The sentiment gap aligns with prior health-communication research: media defaults to cautionary framing while patients share mixed lived experiences. WebMD's strong negativity (-0.31) suggests review-selection bias. The 97.6% classification accuracy—stable under normalisation—confirms fundamentally different registers.

The side-effect coverage gap has real-world implications: patients relying on media may underestimate everyday symptom burden (nausea, constipation, anxiety), while providers could monitor patient communities for emerging signals absent from news. The mental-health aspect discrepancy is particularly striking: media amplifies alarm while patients report only mild negativity.

Limitations. Hybrid mode used 100% snippet fallback (no pre-extracted bodies); VADER may miss medical-domain nuance; 45 weekly observations limit Granger power; Reddit/WebMD are not representative of all patients.

5. Conclusion

Public and media discourse on GLP-1 drugs diverge significantly in sentiment, vocabulary, and thematic emphasis. These differences are robust across analytical methods and length normalisation. Patients relying solely on media may form an incomplete picture; healthcare providers should monitor both streams.

Reproducibility. `python project_cli.py run-analysis --media-text-mode hybrid`